A Ryan Firebee, one of a series of target drones/unpiloted aerial vehicles that first flew in 1951.

quite popular. Around this time, development of underwater unmanned vehicles, or UUVs, also picked up pace gradually. One of the first UUVs, the Special Purpose Underwater Research Vehicle (SPURV), was developed by the University of Washington Applied Physics Laboratory in 1957 with funding from the Office of Naval Research. Even back then, this UUV was capable of diving up to 10,000 feet and could work for four hours.

It was during the Vietnam War that the widespread use of aerial reconnaissance drones normalised UAVs for everyone. Between August 1964 and April 1975, the US would launch about 3,435 Ryan reconnaissance drones over North Vietnam - that's nearly one drone a day for the entire duration of the war. By this time, UUVs had also gathered steam and were mostly used for recovery missions. One underwater drone, the Argo, discovered the wreckage on the Titanic in 1985 and that of Bismarck, the World War II battleship in 1989.

MODERN WARFARE

In the 1980's, Israel developed the Pioneer and the Scout drones, which differentiated themselves from their predecessors with their smaller, lighter, glider type designs - making themselves cheap and harder to shoot down. Among the two, the Scout was especially remarkable in its ability to transmit live 360 degree video of its surroundings. Their effectiveness led to the US purchasing these drones for use in the Gulf War. By the 1990s, US was spending in the billions to develop better UAVs. Around this time, a company known as Leading Systems Inc. was developing a drone named 'Amber' that was designed to endure in the skies. In 1984, DARPA had issued a $40 million contract to the company. Abraham Karem, the owner of Leading Systems, developed the Amber as a drone suited for military operations. However, after a consolidation of UAV efforts, funding for recon programs was cut and the Amber program had to be shut down. At this point, LSI was faced with bankruptcy, and was bought out in 1991 by General Atomics. A few years later, the MQ-1 Predator drone was introduced by General Atomics.

Within 1999, the Predator had been deployed for military purposes and by 2001, it had successfully fired a Hellfire missile. This marked the beginning of an era of drone strikes that were conducted by individuals often sitting on the other side of the globe from the drone, as well as each other. Its successor, the Reaper, was developed with better capabilities.

The MQ-1 Predator

YOU GET A DRONE

While drone development in the military sector has mostly been in line with technology available to the private sector, we didn't see enthusiast drones becoming common until at least 10 years ago. Even then, it wasn't something to be owned by the general public up until very recently. Usage by government agencies and relief efforts began in 2006. That year, FAA in the US granted two commercial drone licenses. Jump to 2015, and that number had risen to a thousand, and 3100 in 2016. If they are to be believed, there will be 600,000 commercial drones in the air by the end of 2018, and that's just the authority that looks after drone regulation, among other things, in one country in the entire world. There's a similar jump in the number of underwater drone projects and budgets set for them. With drone regulations being put into place in India, is that number scheduled to rise here too? More importantly, will they continue being specialised to military usage or will there be drones developed particularly to help civilians? Let's find out in the next cover story.